

Hartmann Sensor Course Notes

1 Light and Wavefronts

Before we build a Hartmann sensor, we need to understand one important idea:

Light is not only a ray. Light is also a wave.

A simple way to imagine a wave is to think about water ripples. If we drop a small stone into water, circular ripples spread outward. Light also has a similar wave-like behavior, although we usually cannot see the wave directly.

1.1 What is a wavefront?

A **wavefront** is a surface connecting points of a light wave that are at the same stage of vibration. In simple words:

Key Idea

A wavefront shows the shape of light as it travels.

The direction of light propagation is usually perpendicular to the wavefront. So if we know the shape of the wavefront, we can understand how the light is moving.

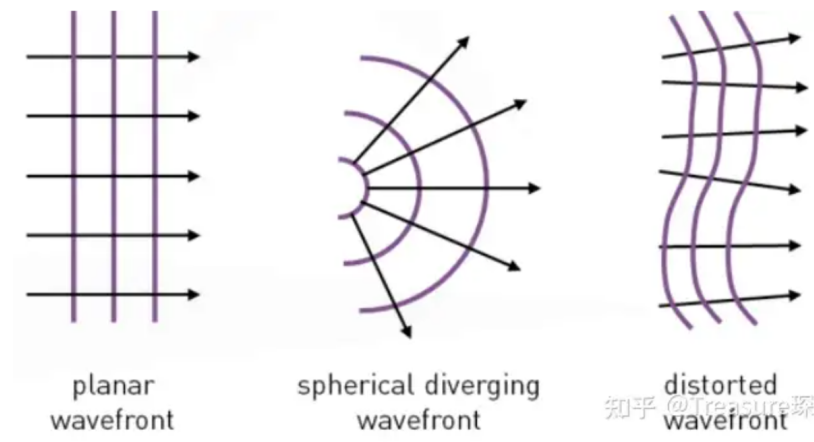


Figure 1: Three examples of wavefronts: planar wavefront, spherical diverging wavefront, and distorted wavefront. The black arrows show the direction of light propagation. The purple curves show the wavefronts.

In a perfect world, light would travel smoothly. But in real optical systems, many things can disturb the wavefront:

- imperfect lenses,
- turbulent air,
- biological tissue,
- misaligned optical components.

After passing through these disturbances, the wavefront is no longer flat or spherical. It becomes a **distorted wavefront**.

1.2 Why do we care about wavefronts?

In imaging systems, the quality of the image is strongly related to the quality of the wavefront.

If the wavefront is flat and smooth, the image can be sharp. If the wavefront is distorted, the image may become blurry.

So our goal is:

Measure the wavefront, understand the distortion, and correct it.

This is the basic idea behind **adaptive optics**.

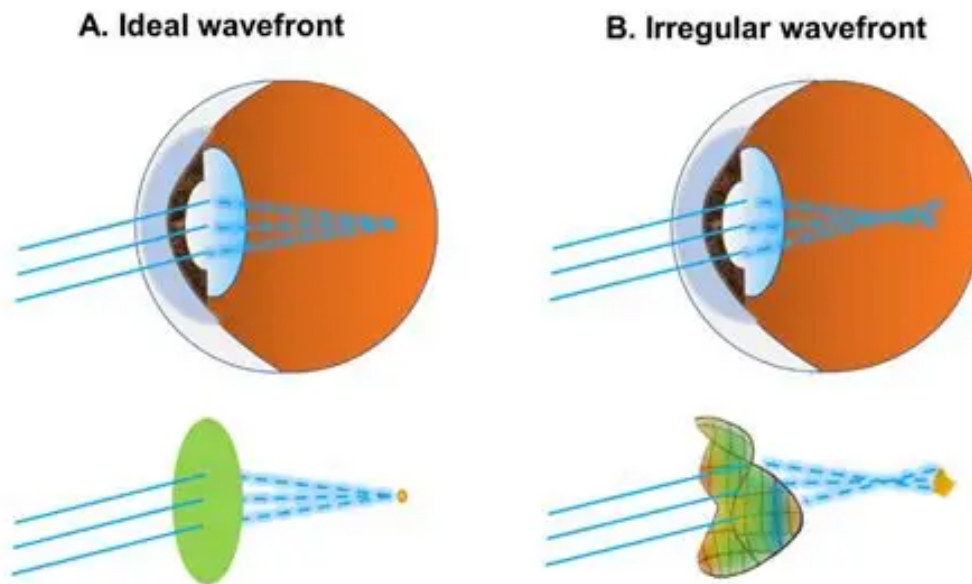


Figure 2: Ideal and irregular wavefronts in an imaging system. In the ideal case, light rays focus well and form a sharp point. In the irregular case, the distorted wavefront prevents the rays from meeting at one perfect focus, so the image becomes blurred.

This is similar to how the human eye forms an image. In a healthy eye, the wavefront is smooth, so light can focus well on the retina and we see a clear image. But if the cornea or lens is not perfect, the wavefront becomes distorted. Then the light cannot focus into a sharp point, and the image becomes blurry.

1.3 A Simple Language for Wavefronts: Zernike Modes

A distorted wavefront can have many different shapes. Some distortions are simple, while others are more complicated.

To describe these shapes, optical scientists often use **Zernike modes**.

A simple way to understand Zernike modes is:

Zernike modes are basic building blocks of wavefront distortion.

Each Zernike mode represents one simple type of wavefront shape. For example:

- **Tilt:** the wavefront is slanted.
- **Defocus:** the focus moves forward or backward.
- **Astigmatism:** the wavefront focuses differently in different directions.
- **Coma:** the wavefront becomes asymmetric, like a comet tail.
- **Spherical aberration:** the center and edge of the wavefront focus differently.

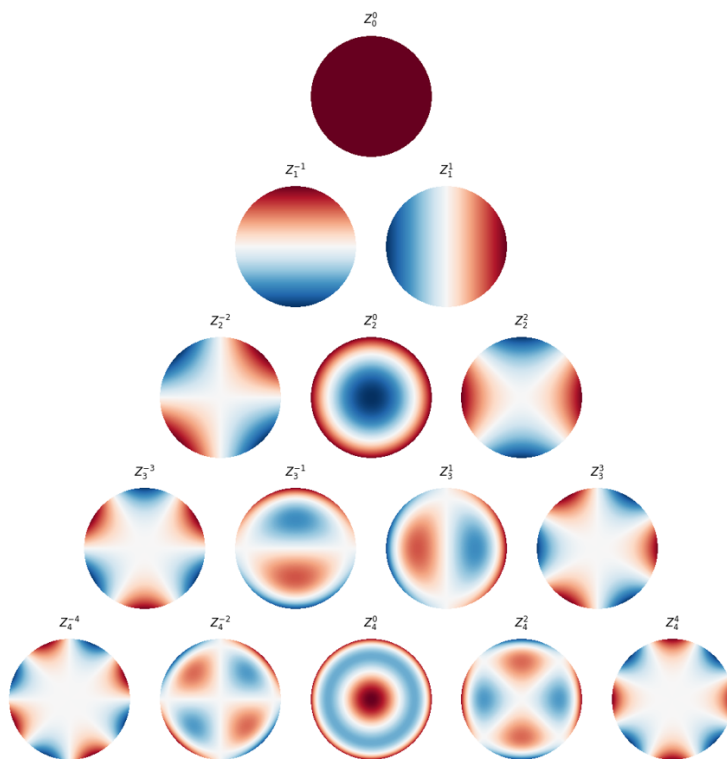


Figure 3: Examples of Zernike modes. Each circular pattern represents one basic type of wavefront shape. By combining many Zernike modes, we can describe a more complicated distorted wavefront.

Mathematically, a wavefront can be written as a sum of Zernike modes:

$$W(\rho, \theta) = \sum_i a_i Z_i(\rho, \theta)$$

Here:

- $W(\rho, \theta)$ is the wavefront shape.
- $Z_i(\rho, \theta)$ is one Zernike mode.
- a_i is the coefficient of that mode.

The coefficient a_i tells us how much of that Zernike mode is present in the wavefront.

Key Idea

Zernike coefficients tell us what kind of distortion exists in the wavefront, and how strong each distortion is.

1.4 From Wavefront to Hartmann Sensor

Now we know that the shape of a wavefront is very important. A smooth wavefront usually gives a sharp image, while a distorted wavefront can make the image blurry.

But there is one problem:

We cannot see the wavefront directly.

So we need a tool to measure it.

The **Hartmann sensor** is a useful tool for this purpose. It helps us detect whether the incoming wavefront is flat, tilted, or distorted.

In this course, we will learn how to build a simple Hartmann sensor and use it to study wavefront distortion.

Main Message

A Hartmann sensor is a tool for measuring wavefront distortion.

2 How Does a Hartmann Sensor Work?

In the previous section, we learned that a distorted wavefront can make an image blurry. Now we need a tool to measure the wavefront.

A **Hartmann sensor** is a simple and powerful tool for this job. It does not measure the whole wavefront directly. Instead, it measures the local direction of light at many small positions.

2.1 The Key Part: A Microlens Array

The most important part of a Hartmann sensor is the **microlens array**. It is a plate with many tiny lenses arranged in a regular grid.

Each small lens is called a **lenslet**. Each lenslet focuses a small part of the incoming light onto a camera.

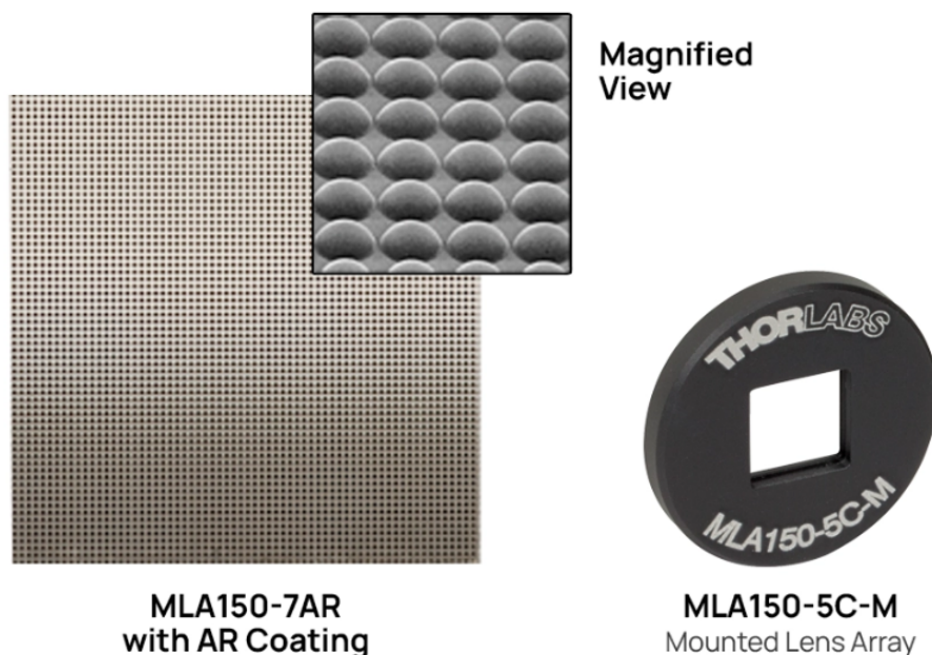


Figure 4: A microlens array. It contains many small lenses. Each small lens can focus one small part of the incoming wavefront.

A simple way to think about it is:

A microlens array cuts one big wavefront into many small pieces.

Each small piece of the wavefront creates one bright spot on the camera.

2.2 From Wavefront Direction to Spot Position

If the incoming wavefront is flat, the light reaches each lenslet straight. Then the spots appear at regular positions on the camera.

If the wavefront is tilted or distorted, the local light direction changes. Then the focused spot moves away from its normal position.

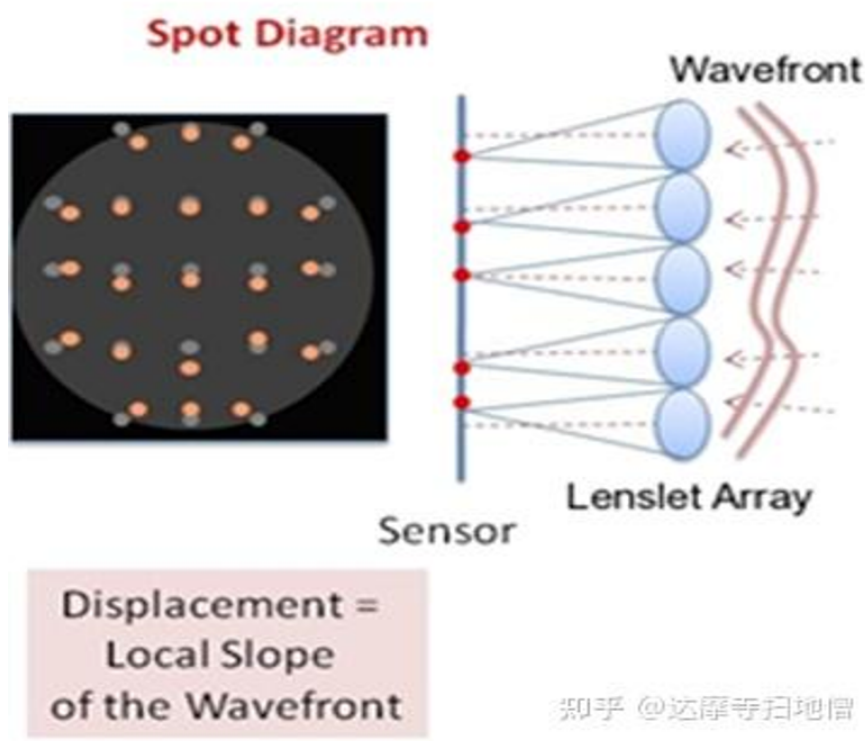


Figure 5: Basic principle of a Hartmann sensor. Each lenslet focuses light onto the sensor. If the local wavefront is tilted, the spot position shifts. The spot displacement tells us the local slope of the wavefront.

This is the central idea:

Key Idea

The movement of each spot tells us the local slope of the wavefront.

2.3 Reference Spots and Distorted Spots

In practice, we usually take two images:

- a **reference image**, measured with a flat or known wavefront;
- a **distorted image**, measured with the unknown wavefront.

For each lenslet, we compare the new spot position with the reference spot position.

If a spot moves to the right, it means the local wavefront is tilted in the x direction. If a spot moves upward or downward, it means the local wavefront is tilted in the y direction.

$$\Delta x = x_{\text{distorted}} - x_{\text{reference}}$$

$$\Delta y = y_{\text{distorted}} - y_{\text{reference}}$$

Here, Δx and Δy are the spot displacements.

2.4 Spot Displacement and Wavefront Slope

The spot displacement is related to the local wavefront slope.

If the focal length of each lenslet is f , then we can write approximately:

$$\frac{\partial W}{\partial x} \approx \frac{\Delta x}{f}$$

$$\frac{\partial W}{\partial y} \approx \frac{\Delta y}{f}$$

Here:

- W is the wavefront shape;
- $\frac{\partial W}{\partial x}$ and $\frac{\partial W}{\partial y}$ are the local slopes of the wavefront;
- Δx and Δy are the measured spot shifts;
- f is the focal length of the microlens.

Main Message

A Hartmann sensor changes an invisible wavefront problem into a visible spot movement problem.

2.5 From Spot Shifts to Wavefront Reconstruction

Now we know that each spot displacement gives us the local slope of the wavefront.

For the k -th lenslet, we measure the spot position in two images:

- reference spot position: (x_k^0, y_k^0)
- distorted spot position: (x_k, y_k)

The spot displacement is

$$\Delta x_k = x_k - x_k^0$$

$$\Delta y_k = y_k - y_k^0$$

If the focal length of the microlens is f , then the local wavefront slopes are approximately

$$s_{x,k} = \frac{\partial W}{\partial x}(x_k, y_k) \approx \frac{\Delta x_k}{f}$$

$$s_{y,k} = \frac{\partial W}{\partial y}(x_k, y_k) \approx \frac{\Delta y_k}{f}$$

Here, $s_{x,k}$ and $s_{y,k}$ are the measured local slopes of the wavefront.

Important Idea

The Hartmann sensor does not directly measure the wavefront height W . It measures the wavefront slope.

2.6 Using Zernike Modes to Describe the Wavefront

To reconstruct the full wavefront, we need a mathematical model.

A common choice is to describe the wavefront as a sum of Zernike modes:

$$W(x, y) = \sum_{j=1}^N a_j Z_j(x, y)$$

Here:

- $W(x, y)$ is the unknown wavefront;
- $Z_j(x, y)$ is the j -th Zernike mode; It is a known function, like a predefined wavefront pattern;
- a_j is the Zernike coefficient;
- N is the number of Zernike modes we use.

The goal of wavefront reconstruction is to find the unknown coefficients

$$a_1, a_2, \dots, a_N$$

Once we know these coefficients, we know the wavefront shape.

2.7 Building the Linear Equations

Because the Hartmann sensor measures slopes, we take derivatives of the Zernike expansion:

$$\frac{\partial W}{\partial x} = \sum_{j=1}^N a_j \frac{\partial Z_j}{\partial x}$$

$$\frac{\partial W}{\partial y} = \sum_{j=1}^N a_j \frac{\partial Z_j}{\partial y}$$

For each lenslet position (x_k, y_k) , we get two equations:

$$s_{x,k} = \sum_{j=1}^N a_j \frac{\partial Z_j}{\partial x}(x_k, y_k)$$

$$s_{y,k} = \sum_{j=1}^N a_j \frac{\partial Z_j}{\partial y}(x_k, y_k)$$

So one lenslet gives two pieces of information:

x-slope and y-slope

If we have many lenslets, we get many equations.

2.8 Matrix Form

All equations can be written in one matrix equation:

$$\mathbf{b} = \mathbf{A}\mathbf{a}$$

where

$$\mathbf{b} = \begin{bmatrix} s_{x,1} \\ s_{y,1} \\ s_{x,2} \\ s_{y,2} \\ \vdots \\ s_{x,K} \\ s_{y,K} \end{bmatrix}$$

is the measured slope vector,

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_N \end{bmatrix}$$

is the unknown Zernike coefficient vector.

The matrix $\mathbf{A}$ contains the derivatives of the Zernike modes at each lenslet position:

$$\mathbf{A} = \begin{bmatrix} \frac{\partial Z_1}{\partial x}(x_1, y_1) & \frac{\partial Z_2}{\partial x}(x_1, y_1) & \cdots & \frac{\partial Z_N}{\partial x}(x_1, y_1) \\ \frac{\partial Z_1}{\partial y}(x_1, y_1) & \frac{\partial Z_2}{\partial y}(x_1, y_1) & \cdots & \frac{\partial Z_N}{\partial y}(x_1, y_1) \\ \frac{\partial Z_1}{\partial x}(x_2, y_2) & \frac{\partial Z_2}{\partial x}(x_2, y_2) & \cdots & \frac{\partial Z_N}{\partial x}(x_2, y_2) \\ \frac{\partial Z_1}{\partial y}(x_2, y_2) & \frac{\partial Z_2}{\partial y}(x_2, y_2) & \cdots & \frac{\partial Z_N}{\partial y}(x_2, y_2) \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix}$$

Here, K is the number of valid lenslets.

2.9 Solving the Equation

Usually, we have more equations than unknowns. So we solve the system using least squares:

$$\hat{\mathbf{a}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$$

This gives the estimated Zernike coefficients.

Then the reconstructed wavefront is

$$\hat{W}(x, y) = \sum_{j=1}^N \hat{a}_j Z_j(x, y)$$

Complete Measurement Mechanism

Spot image $\rightarrow$ spot centers $\rightarrow$ spot displacements $\rightarrow$ local slopes $\rightarrow$ Zernike coefficients $\rightarrow$ reconstructed wavefront.

2.10 A Small Note

The Hartmann sensor measures wavefront slopes, not absolute wavefront height. So the constant part of the wavefront, often called **piston**, cannot be measured from spot shifts.

In most imaging applications, this is not a serious problem, because piston does not change the image sharpness.

2.11 What Do We Finally Get?

After measuring many spot shifts, we know the local slope of the wavefront at many positions.

Then we can reconstruct the full wavefront shape.

So the complete idea is:

Wavefront distortion $\rightarrow$ spot displacement $\rightarrow$ local slope $\rightarrow$ wavefront reconstruction

This is why a Hartmann sensor is very useful in adaptive optics. It tells us how the wavefront is distorted, so that we can later correct it.

3 Adaptive Optics Demo: Optical Path and Correction Idea

In this part, we introduce the optical path used in our adaptive optics demo.

The goal is simple:

Add a wavefront distortion, measure it, and then correct it.

In our setup, the **SLM** is used to control the wavefront. The **Hartmann sensor** is used to measure the wavefront. The camera is used to observe the imaging result.

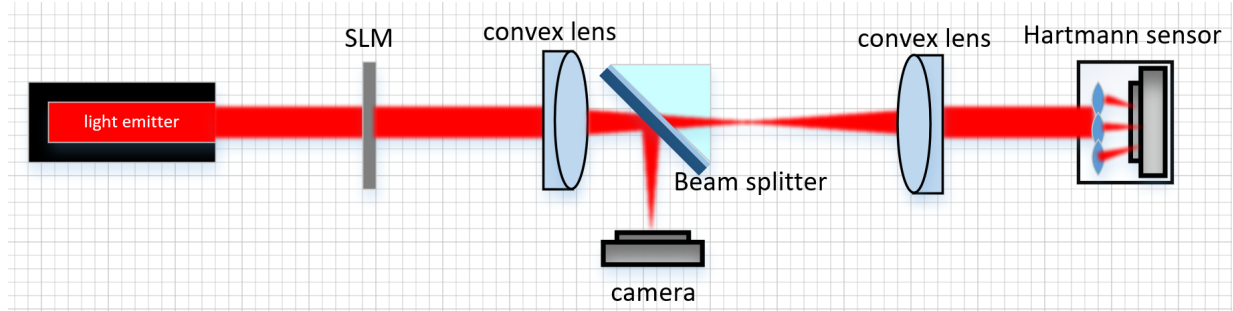


Figure 6: Optical path of the adaptive optics demo. The 635 nm light source passes through the SLM, lenses, beam splitter, camera path, and Hartmann sensor path.

3.1 Main Components

The optical path contains several important parts:

- **635 nm light emitter:** provides the input light.
- **SLM:** adds or corrects the wavefront phase.
- **Convex lenses:** help relay and focus the light.
- **Beam splitter:** divides the light into two paths. One path goes to the camera, and the other path goes to the Hartmann sensor.
- **Camera:** observes the imaging quality.
- **Hartmann sensor:** measures the wavefront distortion.

The most important idea is that the light is divided into two useful paths:

- one path goes to the camera, so we can see the image;
- one path goes to the Hartmann sensor, so we can measure the wavefront.

3.2 Using the SLM to Add a Wavefront

The SLM can change the phase of the light.

In our demo, we first use the SLM to intentionally add a known wavefront distortion. For example, we can add defocus, astigmatism, or coma.

We can write this artificial distortion as

$$W_{\text{add}}(x, y)$$

This means that the SLM is now making the wavefront worse on purpose.

Why do we add distortion on purpose?

We add a known distortion because it gives us a clear test. If our system works, it should be able to measure this distortion and remove it.

3.3 Measuring the Added Wavefront

After the SLM adds the distortion, the distorted light reaches the Hartmann sensor.

The Hartmann sensor records the spot pattern. From the spot shifts, we reconstruct the measured wavefront:

$$W_{\text{meas}}(x, y)$$

Ideally, the measured wavefront should be close to the wavefront added by the SLM:

$$W_{\text{meas}}(x, y) \approx W_{\text{add}}(x, y)$$

This means the Hartmann sensor has successfully detected the distortion.

3.4 Correction: Adding the Opposite Wavefront

Now comes the core idea of adaptive optics.

If the SLM adds a distortion

$$W_{\text{add}}(x, y)$$

then we can correct it by adding the opposite wavefront:

$$W_{\text{corr}}(x, y) = -W_{\text{meas}}(x, y)$$

If the measurement is accurate, then

$$W_{\text{corr}}(x, y) \approx -W_{\text{add}}(x, y)$$

So the total wavefront becomes

$$W_{\text{total}}(x, y) = W_{\text{add}}(x, y) + W_{\text{corr}}(x, y)$$

Substituting the correction term gives

$$W_{\text{total}}(x, y) \approx W_{\text{add}}(x, y) - W_{\text{add}}(x, y) = 0$$

This means the wavefront becomes nearly flat again.

Key Idea

The SLM first adds a bad wavefront. Then we calculate the opposite wavefront and send it back to the SLM. The bad wavefront and the opposite wavefront cancel each other.

3.5 Closed-Loop Correction

In a real experiment, one correction may not be perfect. So we can repeat the process several times.

At each step, the Hartmann sensor measures the remaining wavefront error. Then the SLM applies a new correction.

A simple update rule is

$$W_{\text{corr}}^{(n+1)}(x, y) = W_{\text{corr}}^{(n)}(x, y) - gW_{\text{meas}}^{(n)}(x, y)$$

Here:

- $W_{\text{corr}}^{(n)}$ is the correction wavefront at step n ;
- $W_{\text{meas}}^{(n)}$ is the wavefront measured by the Hartmann sensor;
- g is a small gain factor, usually between 0 and 1.

The gain g controls how strongly we correct the wavefront each time.

If g is too small, the correction is slow. If g is too large, the system may become unstable.

3.6 What Should We See?

Before correction, the wavefront is distorted. The camera image may look blurred or deformed.

After correction, the wavefront becomes flatter. The camera image should become sharper.

So the full adaptive optics loop is:

SLM adds distortion → Hartmann sensor measures it → computer calculates opposite wavefront → SLM applies correction → image becomes better

Main Message

Adaptive optics is a feedback system. It measures the wavefront error and applies the opposite wavefront to reduce that error.